1. Identity

Document	STAR-PKG-STAR-MERCATOR-EN · v2.1 · 2026-11-01
Package	star-mercator
Version	0.1.0
Capability	star.geodesy.mercator
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_mercator

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_mercator-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_mercator-0.1.0-py3-none-any.whl	014398e83071f594ff847235b29b09c86c679f29818581c24df97b29f67d7131
star_mercator.py	427352ea5cee6cebcdc716dce4ed52cf736ae5262d9e21e01c3c54c4f7bc37ee
star_mercator-0.1.0-source.zip	5a28cc589b3d01ffe6ce6f4294c83c3ab107029a06169b6ceac777581aff7709

5. What it does, and its limits

Mercator projection on the ellipsoid with optional latitude of true scale, forward, inverse and scale factor, and Web Mercator; within 6e-8 m of PROJ, 7.5e-9 m of PyGeodesy's isometric latitude and 3e-8 m of both for Web Mercator on 600 cases; reproduces Snyder's published x and scale factor.

Web Mercator is not conformal and its y differs from the ellipsoidal Mercator by tens of kilometres at mid-latitudes: do not mix them. PyGeodesy has no ellipsoidal Mercator: it confirms the isometric latitude (hence y) and Web Mercator, not x or the scale; and it refuses Web Mercator beyond 85.05 deg, so 19 of the 600 cases were compared with PROJ only. The y of Snyder's example is not cited: its last printed digit could not be confirmed. At longitude 180 exactly `web_mercator_forward` returns the East edge (+20037508.34 m) while `mercator_forward` reduces the difference from its central meridian to [-180, 180) and returns the West edge. No rhumb-line distances or bearings.

- `mercator_forward`, `mercator_inverse`: the conformal Mercator of the ellipsoid, y = a·k0·ψ with ψ the isometric latitude written without cancellation, any central meridian, an optional latitude of true scale. Reproduce the x of Snyder's numerical example (USGS PP 1395, p. 266) to the printed 0.1 m.
- `mercator_scale`: point scale factor, 1 on the parallels of true scale. Reproduces Snyder's k = 1.2194146. `web_mercator_forward`, `web_mercator_inverse`: EPSG:3857.
- Measured on 600 cases on WGS-84 (latitudes to ±89, true scale to ±80): forward within 5.5e-8 m and inverse within 6e-8 m of PROJ; y within 7.5e-9 m of PyGeodesy's isometric latitude; Web Mercator within 3e-8 m of PROJ and PyGeodesy; scale within 7.5e-8 of PROJ (whose factors are numerical).
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, latitude beyond ±89.5, latitude of true scale beyond ±80, longitudes beyond ±180, an

ellipsoid outside the stated ranges, plane coordinates beyond the image. Inverse longitudes are in [-180, 180).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_mercator_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-mercator}: Mercator on the ellipsoid, and Web Mercator},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_mercator/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts